Week 4 – Workshop
Gravity: Density, Structures, and Non-Uniqueness

Geophysics 3A: Potential Fields & Geothermics

July 6, 2026

Duration: 3 hours **Setting:** Workshop room (whiteboards; projector)

Preparation

Pre-reading:
Lecture videos:
Notes:

Purpose

This workshop develops intuition about how density varies in porous and compacted materials, how different subsurface structures relate to density contrasts, and how the buried sphere solution illustrates non-uniqueness in gravity. Students work from physical reasoning, analytic solutions, and qualitative cross-section analysis. The session prepares students for Week 5 (filters, upward/downward continuation, reduction-to-the-pole).

Learning Outcomes

- Estimate bulk density of porous materials using simple mixing laws.
- Explain how compaction and pore-fluid substitution change density.
- Predict density contrasts for various geologic structures.
- Interpret gravity anomaly width–depth relationships for the buried sphere.
- Recognise non-uniqueness and explain why lenticular shallow bodies can mimic deep spheres.

Roles and Grouping

Mixed groups of 4–5 with geology and physics/engineering backgrounds. Rotate roles: *Facilitator*, *Math Lead*, *Geology Lead*, *Skeptic/QC*, *Recorder*.

Session Flow (180 min)

0–10 min: Warm-up — "Why does density matter?"

In groups, list three factors controlling density of Earth materials. Share: how do porosity, mineral composition, and pore fluids alter density? (Students recall Week 2 density work with igneous samples.)

10–45 min: Activity A: Density of a Two-Component Porous Sandstone

Goal: Build intuition for porosity, compaction, and fluid-substitution effects.

Tasks (Worksheet A):

1. Derive an expression for the density of a multicomponent system with porosity.
2. Explore how reducing porosity (compaction from burial) and/or changing the type of fluid filling pores affects bulk density.
3. Discuss: which fluid contrast would be measurable with gravity alone?

Groups record 3–4 bullet points about physical intuition (e.g., why air-filled rocks are anomalously light; why compaction increases density nonlinearly).

45–75 min: Activity B: Geologic Structures to Geologic Anomalies

Goal: Connect geology to density contrast and anomaly expectations.

Structures to analyse (Worksheet B):

Group tasks:

1. Discuss geologic structures that may be described by the simple shapes.
2. Propose plausible density contrasts ($\Delta\rho$) given typical lithologies.
3. Predict qualitative anomaly shape: narrow/broad? symmetric/asymmetric? positive/negative?
4. One group presents each structure; whole class discusses similarities and differences.

75–105 min: Activity C: Buried Sphere: Width–Depth and Non-Uniqueness

Goal: Work with the buried sphere analytic solution and explore non-uniqueness.

Given: The vertical gravity from a buried sphere of radius a, density contrast $\Delta\rho$, and centre depth z_0 is:

$$g_z(x) = \frac{4\pi G}{3}\Delta\rho\, a^3 \frac{z_0}{(x^2+z_0^2)^{3/2}}.$$

Tasks (Worksheet C):

1. Measure the *half-width* $x_{1/2}$ from provided plots and derive:
$$\frac{x_{1/2}}{z_0} = \sqrt{2^{2/3}-1}.$$
2. Explain why increasing depth increases width but decreasing $\Delta\rho$ does not.
3. Discuss non-uniqueness: why can shallow, lens-shaped bodies ("lenticular") mimic deep spheres?
4. Instructor demonstration: Python simulation showing buried sphere vs. lenticular bodies with same peak and width.

105–135 min: Superposition: Build-an-Anomaly

Goal: Construct a plausible local cross-section and its gravity by summing components.
Provided: *Superposition Component Library* with scaled kernels (normalized shapes) for: basement step (sedimentary wedge), mafic dike (2D cylinder proxy), felsic pluton (broad polygon), thin regolith cap (negative slab).
Tasks:

1. Draft a geologically plausible cross-section (2–3 bodies) with $\Delta\rho$ from Activity A.
2. Place laterally, set depths/widths, scale amplitudes, and *sum* the responses (trace overlays on graph paper).
3. Annotate why the geology is plausible (lithology, structure, relative timing).

Deliverable: Labeled cross-section + composite gravity + legend of component contributions.

135–165 min: Match-Three: Cross-Sections ↔ Gravity Profiles

Goal: Practice non-uniqueness reasoning with realistic models.
Provided: *Match-Three Handout* — three local-style cross-sections (A–C) and three gravity profiles (1–3) designed to be confusable.
Tasks:

1. Match each profile to the most plausible cross-section using: (i) half-width/FWHM (depth cue); (ii) edge slopes/asymmetry (contacts, wedges); (iii) amplitude scaling with $\Delta\rho$ and size.
2. Propose one additional observation to break any tie (e.g., rock density control, mapping a contact, short-station-spacing gradient, borehole, susceptibility).
3. Role inversion: *Geologists* present the math-based width–depth rationale; *Phys/Eng* defend the geology.

Deliverable: One page with matches (A→?, B→?, C→?) + tie-breaker justification.

165–180 min: Wrap-up

1. Which factor (porosity, fluid, composition) most strongly affects density in shallow basins?
2. Which structural density contrast is hardest to distinguish with gravity alone?

Materials

- Worksheet A: Two-Component Sandstone Density
- Worksheet B: Structure Density Reasoning
- Worksheet C: Buried Sphere and Width–Depth
- Python code for instructor demo

Assessment

- Worksheet submissions (one per group).
- Quality of group discussion and presentations.
- Exit ticket reflection.

Activity A
Density and Porosity

Introduction

In week 2, we calculated several physical properties, though in each case we assumed porosity was negligible. However, for many rocks porosity

Consider a sandstone composed of quartz ($\rho_q = 2650$ kg m^{-3}) and feldspar ($\rho_f = 2600$ kg m^{-3}). The volumetric fractions of solids are f_q and f_f with $f_q + f_f = 1$.

Tasks

1. Write an expression for bulk density:
2. Compute ρ_b for: (i) $\phi = 0.30$, air-filled pores ($\rho_{\text{fluid}} \approx 1$), (ii) $\phi = 0.30$, water-filled pores ($\rho_{\text{fluid}} = 1000$), (iii) $\phi = 0.30$, brine-filled pores ($\rho_{\text{fluid}} = 1200$).
3. Repeat for $\phi = 0.10$ (burial compaction).
4. Explain physically:
   - Why does ρ_b increase with reduced ϕ?
   - Which fluid substitution changes ρ_b the most?

Activity B
Structure Density Reasoning

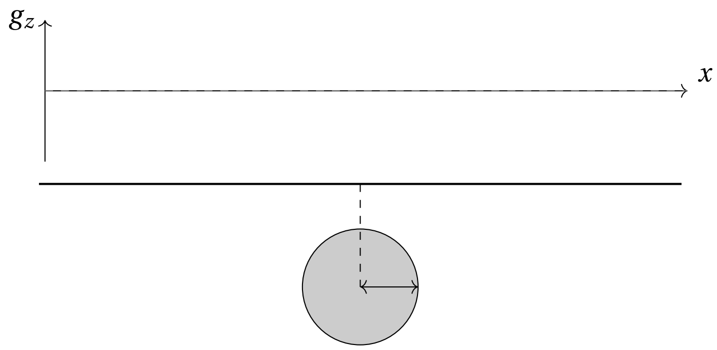
For each assigned structure, complete the table and use the *upper axes* to sketch the expected gravity anomaly $g_z(x)$ (arbitrary units). The *lower panel* shows a schematic cross-section (z positive downward).

Table

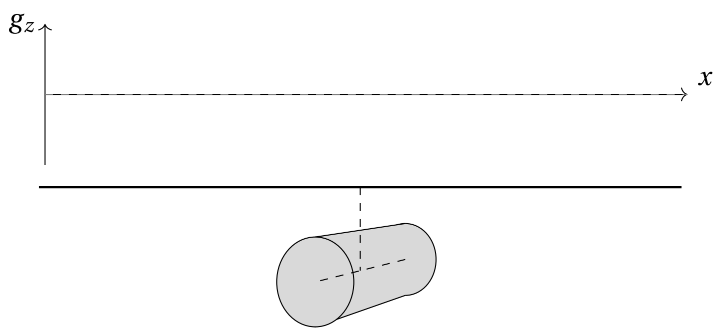
Structure	Likely lithologies	Plausible $\Delta\rho$ ($\mathrm{kg\,m^{-3}}$)
A.		
B.		
C.		
D.		
E.		
F.		
G.		

Prompts: Indicate sign (pos./neg.), width (narrow/broad), symmetry/asymmetry, and where the steepest gradients occur.

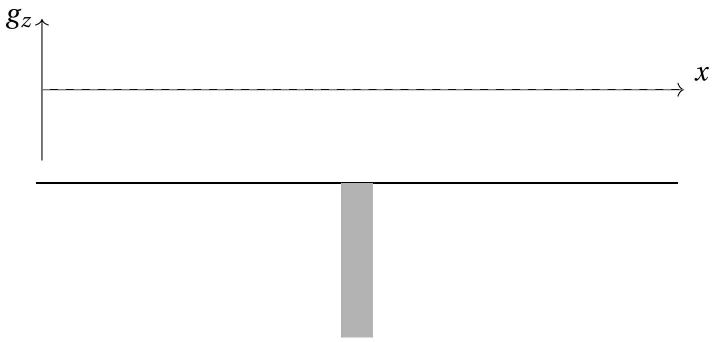
A) Buried Sphere (3D)



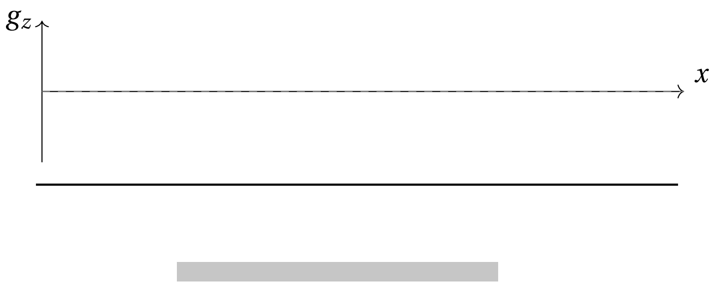
B) Buried Cylinder (2D; infinite strike)



C) Vertical Slab

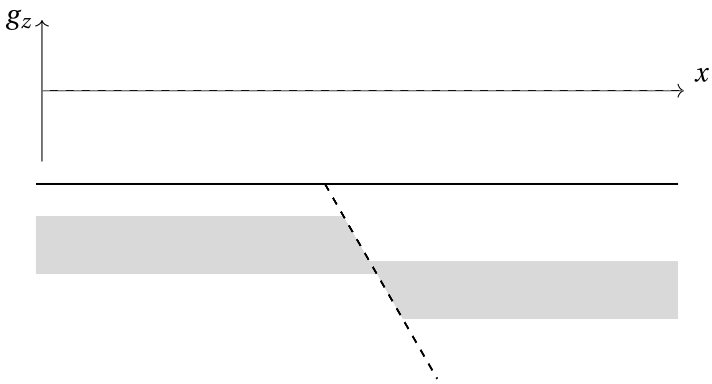


D) Finite-Extent Horizontal Slab

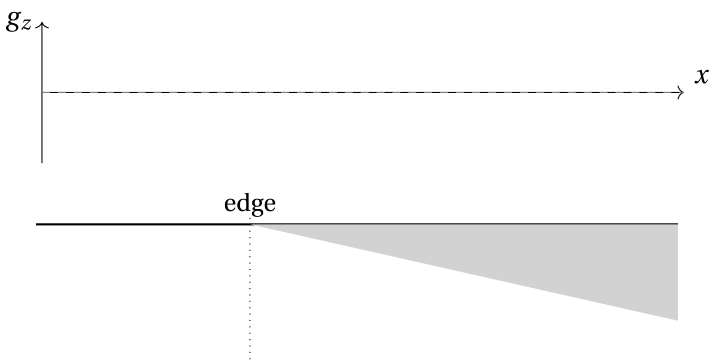


E) Inclined Plane

F) Offset Layer



G) Gently Sloped Basin (edge at one-third from left boundary)



Activity C
Buried Sphere Width–Depth

Given:

$$g_z(x) = \frac{4\pi G}{3} \Delta\rho\, a^3 \frac{z_c}{(x^2 + z_c^2)^{3/2}}.$$

Tasks

1. Show that $g_z(0)$ is proportional to $\Delta\rho\, a^3 / z_c^2$.
2. Solve for $x_{1/2}$ such that $g_z(x_{1/2}) = \frac{1}{2} g_z(0)$.
3. Derive:

$$z_c \approx 0.65\, w$$

   where w is the full-width half-maximum. Hint: this is on the right track,

$$\frac{z_c}{x_{1/2}} = (2^{2/3} - 1)^{-1/2}.$$

4. Explain:
   - Why depth affects width but density contrast does not.
   - How the radius and density contrast result create non-uniqueness for inversion and how we ensure uniqueness.
   - Why a shallow lenticular body could mimic a deep sphere's anomaly.